Permutation and Combination

1. Find the number of ways to arrange the letters of the word 'PENCIL'.

A) 120

B) 240

C) 720

D) 360

Answer: C) 720

2. In how many ways can 5 distinct books be arranged on a shelf?

A) 25

B) 120

C) 24

D) 60

Answer: B) 120

3. Find the value of 7P3.

A) 210

B) 35

C) 840

D) 120

Answer: A) 210

4. Find the value of 10C3.

A) 720

B) 120

C) 240

D) 360

Answer: B) 120

5. In how many ways can a committee of 3 people be chosen from 10 candidates?

A) 120

B) 720

C) 30

D) 1000

Answer: A) 120

6. How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, 5 without repetition?

A) 125

B) 60

C) 20

D) 10

Answer: B) 60

7. How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, 5 if repetition is allowed?

A) 60

B) 125

C) 243

D) 120

Answer: B) 125

8. In how many ways can the letters of the word 'APPLE' be arranged?

A) 120

B) 60

C) 24

D) 12

Answer: B) 60

9. In how many ways can 6 people sit around a circular table?

A) 720

B) 120

C) 60

D) 24

Answer: B) 120

10. How many diagonals does a pentagon have?

A) 5

B) 10

C) 7

D) 6

Answer: A) 5

11. From a group of 7 men and 6 women, 5 persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

A) 564

B) 645

C) 735

D) 756

Answer: D) 756

12. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

A) 360

B) 480

C) 720

D) 5040

Answer: C) 720

13. In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?

A) 810

B) 1440

C) 2880

D) 50400

Answer: D) 50400

14. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

A) 210

B) 1050

C) 25200

D) 21400

Answer: C) 25200

15. In how many ways can a group of 5 men and 2 women be made out of a total of 7 men and 3 women?

A) 63

B) 90

C) 126

D) 45

Answer: A) 63

16. How many 4-letter words with or without meaning can be formed out of the letters of the word 'LOGARITHMS', if repetition of letters is not allowed?

A) 40

B) 400

C) 5040

D) 2520

Answer: C) 5040

17. In how many ways can the letters of the word 'MATHEMATICS' be arranged?

A) 4989600

B) 4989500

C) 4989400

D) 4989300

Answer: A) 4989600

18. A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?

A) 32

B) 48

C) 64

D) 96

Answer: C) 64

19. In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels occupy only the odd positions?

A) 32

B) 48

C) 36

D) 60

Answer: C) 36

20. In how many ways can a cricket team of 11 players be selected out of 16 players if one particular player is always to be included?

A) 3003

B) 4004

C) 5005

D) 6006

Answer: A) 3003

21. In how many ways can a cricket team of 11 players be selected out of 16 players if one particular player is to be excluded?

A) 1365

B) 1820

C) 2400

D) 3003

Answer: A) 1365

22. How many triangles can be formed by joining the vertices of a hexagon?

A) 20

B) 15

C) 10

D) 12

Answer: A) 20

23. Find the number of ways in which 4 boys and 3 girls can be seated in a row so that all the girls sit together.

A) 1440

B) 720

C) 560

D) 240

Answer: B) 720

24. How many numbers between 2000 and 3000 can be formed from the digits 2, 3, 4, 5, 6, 7 when repetition of digits is not allowed?

A) 60

B) 120

C) 30

D) 360

Answer: A) 60

25. In a party, everyone shakes hands with everyone else. If there are 20 persons in the party, find the total number of handshakes.

A) 190

B) 380

C) 400

D) 200

Answer: A) 190

26. There are 10 points in a plane, out of which 4 are collinear. Find the number of lines formed by joining them.

A) 40

B) 39

C) 41

D) 45

Answer: A) 40

27. In how many ways can 5 boys and 5 girls sit in a circle so that no two boys sit together?

A) 5! × 5!

B) 4! × 5!

C) 5! × 4!

D) 4! × 4!

Answer: B) 4! × 5!

28. How many different words can be formed with the letters of the word 'BHARAT'?

A) 360

B) 180

C) 720

D) 120

Answer: A) 360

29. Find the number of permutations of the letters of the word 'ENGINEERING'.

A) 277200

B) 92400

C) 11!

D) 138600

Answer: A) 277200

30. In how many ways can 6 beads of different colors be arranged to form a necklace?

A) 120

B) 60

C) 720

D) 360

Answer: B) 60

31. If nC8 = nC2, find the value of n.

A) 10

B) 12

C) 16

D) 6

Answer: A) 10

32. Find the number of ways of selecting 4 cards from a pack of 52 playing cards.

A) 270725

B) 270750

C) 20725

D) 52!/48!

Answer: A) 270725

33. How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated?

A) 108

B) 120

C) 216

D) 36

Answer: A) 108

34. How many words can be formed from the letters of the word 'SIGNATURE' so that the vowels always come together?

A) 17280

B) 15120

C) 3600

D) 7200

Answer: A) 17280

35. A question paper has two parts, Part A and Part B, each containing 10 questions. If a student has to choose 8 from Part A and 5 from Part B, in how many ways can he choose the questions?

A) 11340

B) 12600

C) 45

D) 11300

Answer: A) 11340

36. How many ways can 7 people be seated at a round table if two particular people must not sit next to each other?

A) 720

B) 480

C) 600

D) 240

Answer: B) 480

37. Find the number of diagonals in a decagon.

A) 35

B) 45

C) 20

D) 90

Answer: A) 35

38. In how many ways can the letters of the word 'DAUGHTER' be arranged so that the vowels never come together?

A) 40320

B) 36000

C) 4320

D) 28800

Answer: B) 36000

39. There are 6 multiple choice questions in an examination. How many sequences of answers are possible, if the first three questions have 4 choices each and the next three have 2 choices each?

A) 512

B) 600

C) 256

D) 128

Answer: A) 512

40. How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67 and no digit appears more than once?

A) 336

B) 330

C) 340

D) 320

Answer: A) 336

41. If 15Pr = 2730, find r.

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

42. How many numbers greater than 1000 can be formed from the digits 0, 1, 2, 3, 4 without repetition?

A) 96

B) 120

C) 192

D) 216

Answer: A) 96

43. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white? (Related concept) Or simply: In how many ways can 2 black and 3 white balls be selected?

A) 840

B) 560

C) 280

D) 120

Answer: A) 840

44. Find the number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines.

A) 6

B) 12

C) 18

D) 9

Answer: C) 18

45. In how many ways can 10 distinct gifts be divided among 3 children so that one receives 5, another 3 and the third 2 gifts?

A) 2520

B) 5040

C) 1260

D) 720

Answer: A) 2520

46. How many numbers divisible by 5 and lying between 3000 and 4000 can be formed from the digits 3, 4, 5, 6, 7, 8 if no digit is repeated in any number?

A) 12

B) 24

C) 6

D) 18

Answer: A) 12

47. In how many ways can a football team of 11 players be selected from 15 players where a particular player must be included?

A) 1001

B) 1365

C) 3003

D) 1000

Answer: A) 1001

48. How many distinct words can be formed using all the letters of the word 'MISSISSIPPI'?

A) 34650

B) 34560

C) 35640

D) 36540

Answer: A) 34650

49. How many 3-digit numbers are there with no digit repeated?

A) 648

B) 729

C) 720

D) 640

Answer: A) 648

50. In how many ways can 5 prizes be distributed among 4 students if each student may get any number of prizes?

A) 1024

B) 625

C) 20

D) 120

Answer: A) 1024

51. Find the number of subsets of a set containing 10 elements.

A) 1024

B) 512

C) 256

D) 100

Answer: A) 1024

52. In how many ways can 8 Indians and 4 Americans be seated in a row so that no two Americans sit together?

A) 8! × 9P4

B) 8! × 9C4

C) 12!

D) 8! × 4!

Answer: A) 8! × 9P4

53. If nP5 = 20 × nP3, find n.

A) 8

B) 9

C) 7

D) 6

Answer: A) 8

54. A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first 5 questions. The number of choices available to him is:

A) 196

B) 280

C) 346

D) 140

Answer: A) 196

55. How many 6-digit numbers can be formed using the digits 0, 1, 3, 5, 7, 9 which are divisible by 10 and no digit is repeated?

A) 120

B) 720

C) 24

D) 60

Answer: A) 120

56. In how many ways can a committee of 5 be formed from 6 men and 4 women so as to include at least 2 women?

A) 186

B) 196

C) 120

D) 160

Answer: A) 186

57. There are 5 different green dyes, 4 different blue dyes and 3 different red dyes. How many combinations of dyes can be chosen taking at least one green and one blue dye?

A) 3720

B) 3100

C) 3255

D) 3600

Answer: A) 3720

58. Find the rank of the word 'MOTHER' if all letters are arranged in alphabetical order.

A) 308

B) 309

C) 310

D) 311

Answer: B) 309

59. In how many ways can 6 persons be selected from 4 officers and 8 constables, if at least one officer is to be included?

A) 896

B) 800

C) 924

D) 224

Answer: A) 896

60. Find the number of ways in which 5 boys and 3 girls can be seated in a row so that no two girls are together.

A) 14400

B) 15600

C) 12000

D) 7200

Answer: A) 14400

61. Ten different letters of an alphabet are given. Words with 5 letters are formed from these given letters. Determine the number of words which have at least one letter repeated.

A) 69760

B) 30240

C) 99748

D) 45360

Answer: A) 69760

62. The number of ways in which a host lady can invite for a party of 5 out of 8 people of whom two do not want to attend the party together is:

A) 56

B) 50

C) 42

D) 28

Answer: C) 42

63. The number of positive integers greater than 6000 and less than 7000 which are divisible by 5, using digits 3, 5, 6, 7, 8 without repetition is:

A) 6

B) 12

C) 24

D) 18

Answer: A) 6

64. How many ways can the letters of the word 'BALLOON' be arranged so that the two L's do not come together?

A) 900

B) 1260

C) 360

D) 600

Answer: A) 900

65. A polygon has 44 diagonals. Find the number of sides.

A) 11

B) 12

C) 10

D) 13

Answer: A) 11

66. In how many ways can 12 books be arranged on a shelf if 4 particular books must always be together?

A) $9! \times 4!$

B) $12!$

C) $8! \times 4!$

D) $9!$

Answer: A) $9! \times 4!$

67. There are 6 pockets in a coat. In how many ways can we put 4 pens in these pockets?

A) 360

B) 1296

C) 4096

D) 24

Answer: B) 1296

68. Find the number of ways in which 7 distinct items can be distributed among 3 persons so that each receives at least one item.

A) 1806

B) 2187

C) 2184

D) 1800

Answer: A) 1806

69. In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?

A) 209

B) 205

C) 194

D) 159

Answer: A) 209

70. How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6, 7 if at least one digit is repeated?

A) 1561

B) 840

C) 2401

D) 1600

Answer: A) 1561

71. From 6 men and 4 women a committee of 4 is to be formed. In how many ways can it be done so that the committee contains at least 1 woman?

A) 195

B) 205

C) 210

D) 185

Answer: A) 195

72. In how many ways can 5 letters be posted in 4 letter boxes?

A) 1024

B) 625

C) 20

D) 120

Answer: A) 1024

73. How many words can be formed with the letters of the word 'PARALLEL' so that all L's are together?

A) 360

B) 720

C) 120

D) 180

Answer: A) 360

74. If 56Pr+6 : 54Pr+3 = 30800 : 1, find r.

A) 41

B) 51

C) 31

D) 21

Answer: A) 41

75. The number of signals that can be made with 5 different flags is:

A) 325

B) 20

C) 120

D) 240

Answer: A) 325

76. In how many ways can a team of 3 boys and 3 girls be selected from 5 boys and 4 girls?

A) 40

B) 20

C) 10

D) 60

Answer: A) 40

77. In how many ways can 5 distinct balls be distributed into 3 distinct boxes?

A) 243

B) 125

C) 15

D) 60

Answer: A) 243

78. How many numbers of 4 digits can be formed with the digits 1, 2, 3, 4, 5, 6 no digit being repeated?

A) 360

B) 340

C) 380

D) 320

Answer: A) 360

79. Find the number of arrangements of the letters of the word 'INDEPENDENCE'.

A) 1663200

B) 3326400

C) 12!

D) 10800

Answer: A) 1663200

80. In how many ways can 6 men and 5 women sit at a round table if no two women are to sit together?

A) $6! \times 5!$

B) 30

C) $5! \times 5!$

D) $7! \times 5!$

Answer: A) $6! \times 5!$

81. How many different signals can be given using any number of flags from 5 flags of different colors?

A) 325

B) 120

C) 240

D) 320

Answer: A) 325

82. In an examination, a student has to answer 4 questions out of 5. The questions are all of multiple choice type with 4 options each. In how many ways can he answer the questions?

A) 1280

B) 256

C) 625

D) 5

Answer: A) 1280

83. The number of ways in which 6 men and 5 women can dine at a round table if no two women are to sit together is given by:

A) $6! \times 5!$

B) 30

C) $5! \times 4!$

D) $7! \times 5!$

Answer: A) $6! \times 5!$

84. In how many ways can 3 prizes be given away to 5 students when each student is eligible for any of the prizes?

A) 125

B) 243

C) 60

D) 15

Answer: A) 125

85. A man has 7 friends. In how many ways can he invite one or more of them to a party?

A) 127

B) 128

C) 64

D) 255

Answer: A) 127

86. How many even numbers of 3 digits can be formed with the digits 0, 1, 2, 3, 4, 5 and 6?

A) 168

B) 180

C) 105

D) 140

Answer: A) 168

87. In how many ways can the letters of the word 'FAILURE' be arranged so that the consonants may occupy only odd positions?

A) 576

B) 144

C) 24

D) 5040

Answer: A) 576

88. If 12Pr = 1320, find r.

A) 3

B) 4

C) 2

D) 5

Answer: A) 3

89. A committee of 5 is to be formed from 9 ladies and 8 men. If the committee commands a majority of ladies, then the number of ways this can be done is:

A) 2412

B) 3400

C) 560

D) 3402

Answer: D) 3402

90. How many words can be formed from the letters of the word 'EXTRA' so that the vowels are never together?

A) 72

B) 48

C) 120

D) 24

Answer: A) 72

91. The number of triangles that can be formed by choosing the vertices from a set of 12 points, 7 of which lie on the same line is:

A) 185

B) 175

C) 115

D) 105

Answer: A) 185

92. In how many ways can 5 boys and 4 girls sit in a row so that the girls occupy the even places?

A) 2880

B) 1440

C) 720

D) 120

Answer: A) 2880

93. Find the number of ways in which 5 distinct books can be distributed among 3 students.

A) 243

B) 125

C) 60

D) 15

Answer: A) 243

94. How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

A) 20

B) 120

C) 24

D) 60

Answer: A) 20

95. In how many ways can a committee of 3 men and 2 women be appointed from a group of 5 men and 4 women?

A) 60

B) 50

C) 40

D) 30

Answer: A) 60

96. How many 3 digit numbers can be formed from the digits 1, 2, 3, 4 and 5 assuming that repetition of digits is allowed?

A) 125

B) 60

C) 120

D) 243

Answer: A) 125

97. Find the value of 15C13.

A) 105

B) 210

C) 15

D) 30

Answer: A) 105

98. In how many ways can 4 red, 3 yellow and 2 green discs be arranged in a row if the discs of the same color are indistinguishable?

A) 1260

B) 362880

C) 120

D) 24

Answer: A) 1260

99. In how many ways can 7 persons be seated at a round table?

A) 720

B) 5040

C) 120

D) 60

Answer: A) 720

100. How many 4-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5 without repetition?

A) 300

B) 360

C) 240

D) 120

Answer: A) 300